

Tensor Analysis

11. Length of a Vector, Angle between Vectors, Geodesics

The quantity $A^p B_p$, which is the inner product of A^p and B_p , is a scalar analogous to the scalar product in rectangular coordinates. We define the length L of the vector A^p or A_4 as given by

$$L^2 = A^p A_p = g^{\rho q} A_p A_q = g_{pq} A^p A^q$$

We can define the angle θ between A^f and B_p as given by

$$\cos \theta = \frac{A^p B_p}{\sqrt{(A^p A_p)(B^p B_p)}}$$

Geodesics

The distance s between two points t_1 and t_2 on a curve $x^r = x^r(t)$ in a Riemannian space is given by

$$s = \int_{t_1}^7 \sqrt{g_{Av^2} \frac{dx^9}{dt} \frac{dx^4}{dt}} dt$$

That curve in the space, which makes the distance a minimum, is called a *geodesic* of the space.

By use of the calculus of variations, the geodesics are found from the differential equation

$$\frac{d^2 x^+}{ds^2} + \left\{ \frac{r}{py} \right\} \frac{dx^v}{ds} \frac{dx^q}{dr} = 0$$

where s is the arc length parameter. As examples, the geodesics on a plane are straight lines whereas the geodesics on a sphere are arcs of great circles.

Physical Components

The physical components of a vector A^p or A_p , denoted by A_s, A_v, A_w are the projections of the vector on the tangents to the coordinate curves are given in the case of orthogonal coordinates by

$$A_{11} = \sqrt{g_{11}} A^1 = \frac{A_1}{\sqrt{g_{11}}}, A_v = \sqrt{g_{vv}} A^2 = \frac{A_2}{\sqrt{g_{vv}}}, A_N = \sqrt{g_{NN}} A^3 = \frac{A_3}{\sqrt{g_{NN}}}$$

Similarly, the physical components of a tensor A^{pq} or A_{pq} are given by

$$A_{wa} = g_{11} A^{11} = \frac{A_{11}}{g_{11}}, A_{5v} = \sqrt{g_{11} g_{22}} A^{12} = \frac{A_{12}}{\sqrt{g_{11} g_{22}}}, A_{un} = \sqrt{g_{11} g_{33}} A^{13} = \frac{A_{13}}{\sqrt{g_{11} g_{33}}}, \text{etc.}$$